

# Performance Testing

AI-ML & GEN-AI Track | Project Phase Documentation

|              |                   |               |                                                          |
|--------------|-------------------|---------------|----------------------------------------------------------|
| Date         | 30 September 2026 | Team ID       | SWTID-2026-8540                                          |
| Project Name | AI BlogNest API   | Track         | AI / ML / Generative AI                                  |
| Team Size    | 5                 | Maximum Marks | 5 Marks                                                  |
| Team Leader  | Pravin kumar .P   | Members       | Abdul Rahaman .A; Dhamotharan .G; Maharaja .C; SUBHASH S |

## Performance and Reliability Testing Plan

| Test Area                 | Method / Check                                                                    | Expected Result                                                              | Evidence to Capture                             |
|---------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------|
| API Response Time         | Run repeated requests for common read/create routes in Postman or Thunder Client. | Stable response times without unnecessary variability.                       | Response time screenshots / exported collection |
| AI Generation             | Send valid and invalid prompts to the generation endpoint.                        | Valid requests return generated content; invalid input returns clear errors. | Request/response evidence                       |
| Summarization             | Submit short and long articles.                                                   | Summary remains concise and service handles limits safely.                   | Input/output samples                            |
| Authentication            | Test valid, expired and malformed JWTs.                                           | Only valid tokens reach protected routes.                                    | HTTP status results                             |
| RBAC                      | Attempt actions as Admin, Editor, Author and Reader.                              | Each role receives only its permitted actions.                               | Role test matrix                                |
| Validation / Sanitization | Send missing, malformed and suspicious parameters.                                | Invalid data is rejected and unsafe input is sanitized.                      | Negative-test evidence                          |
| Rate Limiting             | Repeat requests above configured threshold.                                       | Excess traffic receives controlled rejection.                                | Rate-limit response evidence                    |

## Acceptance principle

A test is considered successful when the API returns the expected status, response structure and authorization behavior without exposing secrets or internal stack traces.